

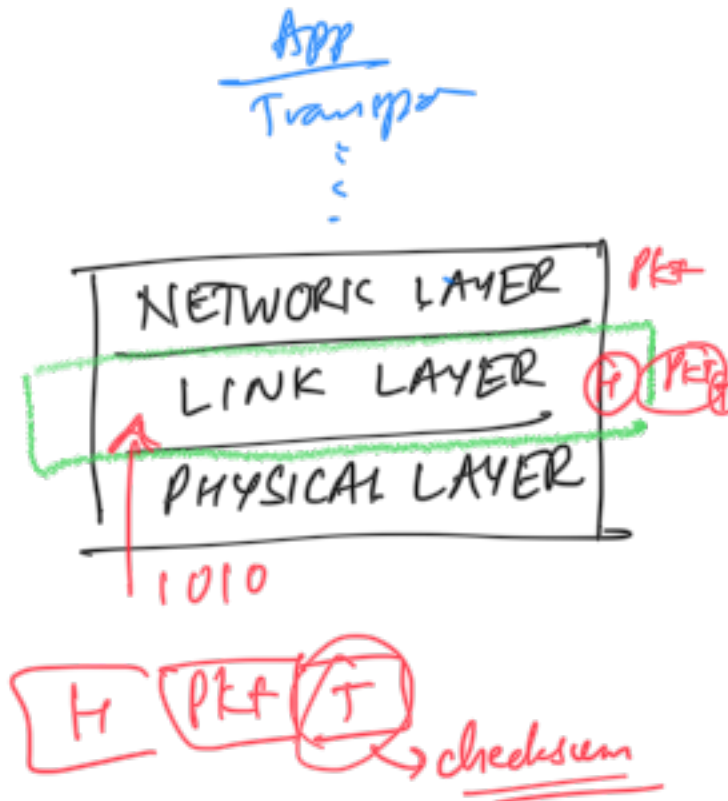
Lec 26A: Link Layer Error detection & correction

21/11/23

* Link Layer services (frame)

- error detection
- error correction (optional)
- flow control
- access control
- framing & synchronization

provides services to network layer



ERROR DETECTION & CORRECTION

Parity checking.

- single bit parity
- 2-dimensional bit parity (refer slides)

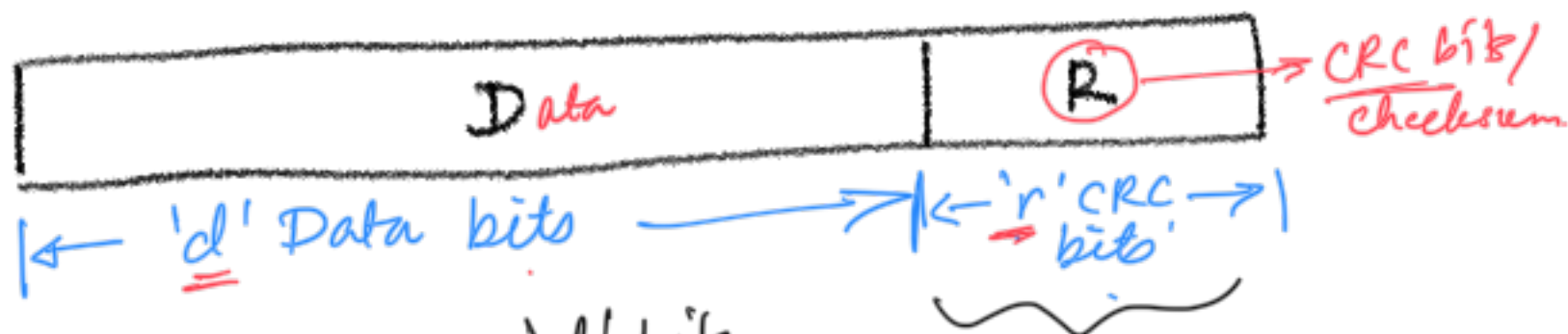
Internet checksum

- error detection
- implemented by TCP, UDP, IP, ...

Cyclic Redundancy Check (CRC)

- powerful error detection & correction
- implemented by Ethernet, ...

* Cyclic Redundancy Check (CRC)



D: Data bits $\Rightarrow$ 'd' bits $\leftarrow$

R: CRC bits $\Rightarrow$ 'r' bits $\leftarrow$

G: Generator bit pattern $\Rightarrow$ 'r+1' bits $\leftarrow$ Generator polynomial

* can detect burst errors less than r+1 bits

At the sender: Generator, G $\Rightarrow$ Public

$$\langle D, R \rangle = \underline{D} * 2^r \text{ XOR } R$$

(1010... * 2^r)



$$\text{Remainder} \left[\frac{D * 2^r}{G} \right]$$

modulo 2 division

$$G = x^2 + x + 1$$

$$G = 1011 \Rightarrow 'r+1'$$

4 bits

$$r = 3$$

Eg. $D = 101110$ $d = 6 \text{ bits}$
 $G = 1001$ $r+1 = 4 \text{ bits} ; r=3$

$$\langle D, R \rangle = D * 2^r + \text{Rem} \left(\frac{D * 2^r}{G} \right)$$

(r=3) $\hookrightarrow$ modulo 2 division

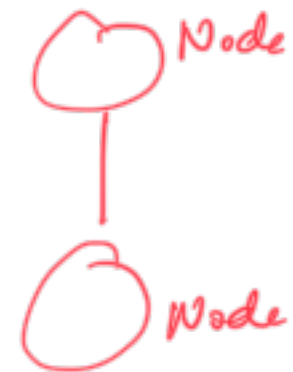
$$D * 2^r = 101110000$$

$$\begin{array}{r}
 101011 \\
 1001 \overline{) 101110000} \\
 \underline{\oplus 1001} \downarrow \downarrow \downarrow \downarrow \\
 001010 \downarrow \downarrow \downarrow \downarrow \\
 \underline{1001} \downarrow \downarrow \downarrow \downarrow \\
 001100 \downarrow \downarrow \downarrow \downarrow \\
 \underline{1001} \downarrow \downarrow \downarrow \downarrow \\
 01010 \downarrow \downarrow \downarrow \downarrow \\
 \underline{1001} \downarrow \downarrow \downarrow \downarrow \\
 0011
 \end{array}$$

$$R = 011$$

$$\langle DR \rangle \quad 101110011$$

$\underbrace{\hspace{1.5cm}}_D \quad \underbrace{\hspace{1.5cm}}_R$



Receiver receives $\langle RD \rangle$

* Compute $\text{Rem}(RD/G)$

$$\hookrightarrow \text{Rem}(RD/G) = \underline{0} \Rightarrow \text{no error}$$

$\hookrightarrow \text{Rem}(RD/G) \neq 0 \Rightarrow \text{error is detected;}$
 drop the frame
 or
 ask for retransmission